A Doughnut-Economics Constraint Model of Floor and Ceiling Trade-offs: An Answer Set Programming Approach

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1. Introduction and problem statement

Kate Raworth's *Doughnut Economics* (2017) reframes the goal of an economy as staying within a "safe and just space": above a **social floor** that guarantees everyone the essentials of a decent life, and below an **ecological ceiling** defined by the planet's biophysical limits. The framework is intuitively appealing and has been adopted by cities such as Amsterdam as a policy compass, but it is rarely turned into a *formal reasoning problem*: given a country's current situation, does a combination of feasible policies exist that closes every social gap and respects every ecological boundary at once, and if not, why not?

This project treats that question as a **constraint satisfaction problem**, encoded in **Answer Set Programming (ASP)** and solved with `clingo`. Given a country's baseline on 7 ecological and 11 social indicators, and a menu of 14 candidate policy levers each with an estimated effect on one or more of them, the model searches for the most parsimonious lever subset that brings every indicator inside the doughnut. When no such subset exists, it does not just report failure: it computes the minimal remaining violation and identifies which dimensions stay structurally out of reach, which serves as a computational explanation of the impossibility.

ASP suits a computational-ethics context better than a generic linear or integer program would, because it represents the problem declaratively, as facts and constraints close to the natural-language statement of the two ethical principles at stake ("no dimension may be below the floor," "no dimension may exceed the ceiling"), rather than as a bespoke optimization routine. That same declarative form also makes the model easy to extend, since a new lever or country is just a new fact, and easy to interrogate, since it is always possible to ask which constraints fail and under which assumptions.

2. Ethical framework

The model formalizes two distinct normative theories, one governing each side of the doughnut:

- **Sufficientarianism** (Frankfurt, 1987; the capability approach of Sen, 1999, and Nussbaum, 2011) governs the social floor. Its central claim is that what matters morally is that everyone reaches a sufficient level of capability/well-being; falling short on *any* dimension is a violation that cannot be offset by surplus on another dimension. This is why the model treats each of the 11 social indicators as an independent hard constraint rather than averaging them into a single welfare score: an average could hide a severe deficit in, say, sanitation behind a surplus in income, which sufficientarianism explicitly rules out as acceptable.
- **Strong sustainability**, grounded in the planetary boundaries framework (Rockström et al., 2009), governs the ecological ceiling. Unlike "weak" sustainability, which allows trading natural capital for produced or economic capital, strong sustainability treats certain biophysical boundaries as non-substitutable: no amount of economic benefit justifies crossing them. This is why the 7 ecological indicators are also independent hard constraints, not a single weighted ecological index.

The combination of the two is what makes the "doughnut" shape: a space bounded by two structurally different kinds of constraint, justified by two different ethical theories, both of which reject the more common move of building a single aggregate objective function. This is worth stating explicitly because it is the modeling decision most exposed to challenge in discussion. A single aggregate, such as a weighted sum of all 18 indicators, would have been computationally simpler, but ethically wrong under either theory.

3. Data

All baseline country data come from O'Neill, Fanning, Lamb & Steinberger (2018), *"A good life for all within planetary boundaries"*, Nature Sustainability 1, 88–95. The paper computes, for 151 countries, 7 biophysical indicators (CO_2 emissions, phosphorus, nitrogen, blue water, embodied HANPP, ecological footprint, material footprint) and 11 social indicators (life satisfaction, healthy life expectancy, nutrition, sanitation, income, access to energy, education, social support, democratic quality, equality, employment), each expressed as a **ratio to a threshold**:

- Biophysical indicators: value ≤ 1.0 means the per-capita share of the planetary boundary is respected; value > 1.0 means overshoot.
- Social indicators: value ≥ 1.0 means the threshold for a good life is met; value < 1.0 means a deficit.

The official supplementary data file (`41893_2018_21_MOESM2_ESM.xlsx` , hosted by Springer Nature) was downloaded directly and converted into ASP facts by `src/data_prep.py` , scaling every ratio by 100 and rounding to the nearest integer so the encoding can use integer arithmetic (`clingo` has no native floating-point support). For example, Vietnam's CO2 ratio of 1.35 is stored as the fact `bio_value(vietnam, co2, 135)` .

This is real, published, peer-reviewed data rather than a synthetic scenario, so the trade-offs the model surfaces are trade-offs a real country actually faces, not artifacts of an invented example.

4. Formalization

Turning the doughnut into something a solver can actually check means building a short pipeline, each stage handling one part of the problem:

1. **`data_prep.py`** turns the raw O'Neill et al. spreadsheet into ASP facts describing where a country currently stands (Section 4.1).
2. **`levers.lp`** declares the menu of 14 candidate policies and their estimated effects (Section 4.2).
3. **`doughnut.lp`** is the core ASP program: it chooses a subset of levers and checks whether it satisfies every floor and ceiling constraint at once (Section 4.3).
4. **`best_effort.lp`** takes over when no subset works, diagnosing how far off the best achievable outcome still is (Section 4.4).
5. **`run_scenario.py`** drives `clingo` end to end and writes the results discussed in Section 7.
6. **`representativeness_check.py`** independently checks that the lever effects assumed in step 2 are realistic (Section 5).

The rest of this section walks through the two ASP programs, `doughnut.lp` and `best_effort.lp`, block by block. Section 6 covers how to actually run this pipeline.

4.1 Facts

For each country `C` , dimension `D` , and baseline value `V` :

```
 bio_value(C, D, V).   % D in {co2, phosphorus, nitrogen, blue_water,
                      %       ehanpp, eco_footprint, material_footprint}
soc_value(C, D, V).   % D in {life_satisfaction, healthy_life_exp, ...}
```

4.2 Levers and their effects

Each policy lever `L` is declared once, together with the dimensions it affects and by how much (a signed integer delta, in the same x100 scale):

```
 lever(L).
affects_bio(L, Dim, Delta).
affects_soc(L, Dim, Delta).
```

A lever is a binary intervention: at solve time it is either selected or not (Section 4.3), and if selected it contributes a fixed, pre-declared delta to every dimension it affects. The solver never decides how much of a lever to apply, only whether to apply it at all. A lever can touch several dimensions at once, and two levers can be declared mutually exclusive via `conflicts/2` (used once, for `l1` / `l11` below). All 14 levers, declared in `src/levers.lp`, are:

#	Lever	What it represents	Effect on ecological dimensions	Effect on social dimensions	Indicative source (direction/order of magnitude only)
L1	`l1_renewable_energy`	Renewable-energy transition (wind/solar/hydro investment displacing fossil generation)	co2 −35	energy_access +15	IPCC AR6 WGIII mitigation pathways; IEA net-zero scenario
L2	`l2_circular_economy`	Circular-economy / material policy (reuse, recycling, product-life extension)	material_footprint −30; eco_footprint −15	—	Ellen MacArthur Foundation circular-economy reports
L3	`l3_sustainable_agriculture`	Sustainable/regenerative agriculture (lower nutrient runoff and land pressure)	nitrogen −25; phosphorus −25; ehanpp −30	nutrition −10 (modeled trade-off: lower-intensity farming reduces short-term nutritional surplus)	FAO / EAT-Lancet on regenerative agriculture & nutrient runoff
L4	`l4_water_efficiency`	Water-efficiency infrastructure and demand management	blue_water −20	—	UN water-efficiency reports
L5	`l5_working_time_reduction`	Working-time reduction (e.g. four-day week)	eco_footprint −20; material_footprint −15	life_satisfaction +22; employment +10	Kallis et al. (degrowth); four-day-week trials (Iceland, Autonomy UK)

#	Lever	What it represents	Effect on ecological dimensions	Effect on social dimensions	Indicative source (direction/order of magnitude only)
L6	`l6_universal_basic_income`	Universal basic income	—	income +20; sanitation +15; nutrition +10	ILO Social Protection Floor Recommendation 202
L7	`l7_universal_healthcare`	Universal healthcare coverage	—	healthy_life_exp +20; social_support +10	WHO Universal Health Coverage reports
L8	`l8_participatory_governance`	Participatory governance (participatory budgeting, deliberative bodies)	—	democratic_quality +30; social_support +10	OECD participatory-budgeting / open-government studies
L9	`l9_wealth_tax`	Progressive wealth taxation	—	equality +25; income +10	Piketty, *Capital in the Twenty-First Century*
L10	`l10_education_investment`	Public education investment	—	education +20; employment +10	UNESCO education-investment reports
L11	`l11_fossil_energy_expansion`	Fossil-fuel-based energy access expansion, cheaper and faster than L1, but dirtier; `conflicts(l1, l11)`	co2 +15	energy_access +20	Deliberately modeled as the cheap/dirty alternative to L1 (Section 4.3)
L12	`l12_decentralization_reform`	Decentralization / subsidiarity reform	—	democratic_quality +20; social_support +5	Subsidiarity/ decentralization literature (Ostrom)
L13	`l13_sanitation_infrastructure`	Sanitation infrastructure investment	—	sanitation +25	WHO/UNICEF WASH programme reports
L14	`l14_carbon_fee_dividend_reforestation`	Carbon fee-and-dividend combined with large-scale reforestation	co2 −50; ehanpp −35	equality +15 (fee revenue returned as an equal per-capita dividend)	Hansen et al. 2013 (fee-and-dividend); IPCC AR6 WGIII (reforestation as a carbon sink)

As Section 5 discusses, every delta in this table is an illustrative assumption about direction and rough order of magnitude, not a fitted causal estimate: the "indicative source" column names the literature that motivates the sign and ballpark of each effect, not a regression this project reproduces.

4.3 Choice and constraints

The core of the encoding (`src/doughnut.lp`) is a **choice rule** over the lever menu, followed by hard integrity constraints that encode the two ethical principles from Section 2:

```
{ select(L) } :- lever(L).

:- select(L1), select(L2), conflicts(L1, L2).

bio_effect(Dim, Sum) :-
    bio_dim(Dim),
    Sum = #sum { Delta,L : select(L), affects_bio(L, Dim, Delta) }.

soc_effect(Dim, Sum) :-
    soc_dim(Dim),
    Sum = #sum { Delta,L : select(L), affects_soc(L, Dim, Delta) }.

new_bio_value(C, Dim, V + E) :- target(C), bio_value(C, Dim, V), bio_effect(Dim, E).
new_soc_value(C, Dim, V + E) :- target(C), soc_value(C, Dim, V), soc_effect(Dim, E).

ceiling_ok(C, Dim) :- new_bio_value(C, Dim, V), V <= 100.
floor_met(C, Dim)  :- new_soc_value(C, Dim, V), V >= 100.

:- target(C), bio_value(C, Dim, _), not ceiling_ok(C, Dim).
:- target(C), soc_value(C, Dim, _), not floor_met(C, Dim).

#minimize { 1,L : select(L) }.
```

A rule of the form `head :- body.` means "if body is true, derive head." A rule with no head, `:- body.`, is instead an integrity constraint: it forbids the body from ever being true, discarding the whole candidate answer set rather than penalizing it. `{ select(L) } :- lever(L).` is the opposite kind of rule, a choice rule marked by the curly braces: the solver is free to make each `select(L)` true or false independently, generating up to 2^14 candidate lever subsets as the starting search space, and everything below only filters or scores that space. `conflicts/2` puts the integrity-constraint pattern to its one real use here, ruling out `l1_renewable_energy` and `l11_fossil_energy_expansion` together, since the two levers reach the same energy-access target through opposite ecological means: forcing a choice between them gives the model a genuine normative fork, where which one a solution uses is settled by which side effects are considered acceptable rather than by the formalism itself, precisely the kind of decision computational ethics is meant to expose rather than paper over.

The two `#sum` rules add up the deltas of every selected lever on each dimension, including the lever identifier `L` inside the aggregate's tuple so that two different levers with the same delta on the same dimension are not silently collapsed into one (this additive, no-interaction simplification is revisited as a limitation in Section 8). `new_bio_value` and `new_soc_value` add that net effect to the baseline of whichever country is the current `target(C)`, a fact injected by `run_scenario.py` at solve time so the same program serves every country unchanged. `ceiling_ok` and `floor_met` just name whether a dimension passes, and the two constraints right after them enforce it one dimension at a time: because nothing is folded into a single combined score, a solution that fixes every dimension but one is still thrown out entirely, which is exactly the non-substitutability that sufficientarianism and strong sustainability demand (Section 2).

Among the answer sets that survive, `#minimize { 1,L : select(L) }` prefers those using the fewest levers, a parsimony criterion that is itself a small normative assumption made explicit here rather than left implicit. `run_scenario.py` calls `clingo` with `--opt-mode=optN`, which keeps every answer set tied at this optimal cost instead of stopping at the first one found, and that is what surfaces cases like Costa Rica's four equally parsimonious solutions in Section 7.2.

4.4 The diagnostic model for unsatisfiable cases

When no subset of levers satisfies every constraint, `doughnut.lp` correctly returns UNSAT, a real result since the problem is genuinely infeasible under the model's assumptions, but simply reporting that would leave Section 7 with nothing more to say about Costa Rica or the United States than "no," which is not useful for either a policy discussion or an ethical argument. `src/best_effort.lp` turns that "no" into a measurement, reusing the same facts, lever menu and effect-aggregation machinery as `doughnut.lp` but dropping the two hard constraints, and instead minimizing how much of the floor and ceiling is still violated once the best available combination of levers has been applied.

```
 overshoot(Dim, V - 100) :- new_bio_value(_, Dim, V), V > 100.
deficit(Dim, 100 - V)   :- new_soc_value(_, Dim, V), V < 100.

#minimize { P,Dim : overshoot(Dim, P) }.
#minimize { P,Dim : deficit(Dim, P) }.
```

The two `#minimize` statements carry no explicit priority, so `clingo` treats them as one joint objective, and the optimal model reports a single "total residual violation" figure rather than overshoot and deficit scored separately. This is exactly the number Section 7 uses to compare Costa Rica's pre-fix outcome (77) against the United States (2815), and the gap between them, roughly two orders of magnitude, is the more robust part of the argument in Section 8: even if the exact lever effect sizes were debated, that ordering is unlikely to flip. Selecting no levers at all is always a valid, if poor, answer set, so this diagnostic model is always satisfiable: it plays the role a minimal unsatisfiable core would play in a purely logical setting, but expressed in units an ethical argument can actually use rather than a list of conflicting rule instances (a deliberate substitution revisited as a limitation in Section 8), and `run_scenario.py` only calls on it once `doughnut.lp` itself has already come back UNSAT.

5. Representativeness of the assumed lever effects

The 14 levers in Section 4.2 are illustrative rather than fitted: there is no dataset of causal, per-country policy effects for these indicators, so every effect size is a simplified estimate of plausible direction and rough size, not a parameter estimated from data. Stating this as a limitation is not enough on its own, so `src/representativeness_check.py` makes the assumption checkable, comparing every assumed effect against how much the same indicator actually varies across the 151 real countries in the O'Neill et al. dataset.

Concretely, for each indicator the script looks at the full distribution of values across all 151 countries and asks two questions. First, how spread out are the middle half of countries, from the 25th to the 75th percentile? This is the interquartile range, or IQR, and it captures the kind of variation that is common and unremarkable: most countries differ from each other by roughly this much. Second, how far apart are a fairly typical low country and a fairly typical high country, from the 10th to the 90th percentile? This wider P10–P90 range captures variation that is still real and observed, but closer to the extremes. An assumed lever effect is judged representative if it stays within the kind of gap that already exists between real countries, so that it does not ask a country to jump further than countries already differ from one another.

The result, tabulated in full in `report/representativeness_table.md`, is that no lever's assumed effect exceeds the real P10–P90 spread on its dimension. Most levers are conservative by this standard, using well under half of even the narrower IQR: the renewable-energy lever's assumed −35 point effect on CO_2, for instance, is only 7% of the real IQR. A smaller group of levers is more ambitious, using between 45% and 89% of the IQR: universal basic income on income, the wealth tax on equality, participatory governance and decentralization reform on democratic quality, universal healthcare on healthy life expectancy, working-time reduction on life satisfaction, and the carbon-fee-and-dividend lever's effects on equality and eHANPP. These are flagged explicitly, since they are the assumptions most likely to be challenged in discussion, though even they stay smaller than a difference that already exists between two comparable real countries.

6. Running the pipeline

All code is Python 3.11 and `clingo` 5.8.0, isolated in a project-local virtual environment with a two-line `requirements.txt` (`clingo`, `openpyxl`). After `pip install -r requirements.txt`, the whole pipeline described in Section 4 runs with three commands:

```
python3 src/data_prep.py                        # spreadsheet -> ASP facts
python3 src/representativeness_check.py        # checks lever effects (Section 5)
python3 src/run_scenario.py <country_slug>      # e.g. vietnam, costa_rica, united_states
```

`data_prep.py` and `representativeness_check.py` only need to be run once; `run_scenario.py` is run once per country and writes its result to `results/<slug>.json`, which is exactly what Section 7's tables are built from.

Two implementation details are worth knowing if a number looks surprising when reproducing these results. `data_prep.py` skips a spreadsheet cell rather than zero-filling it when it is blank, which is why Vietnam ends up with only 17 of the usual 18 tracked dimensions (Section 7.1): a gap inherited from the source data, not a bug in this pipeline. `run_scenario.py` asks `clingo` for every answer set tied at the optimal cost, not just the first one found, which is why Costa Rica comes back with four equally parsimonious solutions instead of one (Section 7.2); a small de-duplication step then filters out the occasional duplicate `clingo` produces when two models differ only on atoms that are not shown.

7. Case studies and results

Three countries were selected after ranking all 151 by worst ecological overshoot and worst social deficit, to cover a spectrum of difficulty: Vietnam (mild overshoot on a single ecological dimension, moderate social gaps), Costa Rica (overshoot on six of seven ecological dimensions, but only two small social deficits), and the United States (overshoot on *all seven* ecological dimensions, some by an order of magnitude, but nearly all social dimensions already met).

7.1 Vietnam: feasible, with equivalent alternatives

Vietnam is missing the `education` indicator in the source spreadsheet (Section 6), so only 17 of the usual 18 dimensions are tracked for this country. In the baseline, CO2 is the only ecological violation, at 135, and four social dimensions fall short: democratic quality at 52, sanitation at 76, equality at 77, and life satisfaction at 79.

(Values are x100 ratios-to-threshold; ceiling dimensions pass at ≤100, floor dimensions pass at ≥100. Bold = changed by the selected levers.)

Dimension	Type	Baseline	Model 1 (via L1)	Model 2 (via L14)
co2	ceiling	135	**100**	**85**
phosphorus	ceiling	84	84	84
nitrogen	ceiling	88	88	88
blue_water	ceiling	38	38	38
ehanpp	ceiling	73	73	**38**
eco_footprint	ceiling	79	**59**	**59**
material_footprint	ceiling	92	**77**	**77**
life_satisfaction	floor	79	**101**	**101**
healthy_life_exp	floor	102	102	102
nutrition	floor	100	100	100
sanitation	floor	76	**101**	**101**
income	floor	101	**111**	**111**
energy_access	floor	105	**120**	105
social_support	floor	100	**115**	**115**
democratic_quality	floor	52	**102**	**102**
equality	floor	77	**102**	**117**
employment	floor	116	**126**	**126**

The problem turns out to be satisfiable, with two equally parsimonious optimal models of six levers each, differing only in how CO2 is closed:

- **Model 1:** `l1_renewable_energy, l5_working_time_reduction, l8_participatory_governance, l9_wealth_tax, l12_decentralization_reform, l13_sanitation_infrastructure`
- **Model 2:** the same, with `l1_renewable_energy` swapped for `l14_carbon_fee_dividend_reforestation`

Both are formally equivalent ways to satisfy the CO2 ceiling. L14 additionally closes `ehanpp` further and lifts `equality` higher, through its per-capita dividend, but leaves `energy_access` at its baseline since, unlike L1, it does not target that dimension. All 17 tracked dimensions land inside the doughnut in both models.

7.2 Costa Rica: infeasible until a more ambitious lever is added

In the baseline, six of the seven ecological dimensions are in overshoot: CO2 at 172, phosphorus at 120, nitrogen at 114, eHANPP at 157, ecological footprint at 129, material footprint at 141. Only equality (62) and employment (93) fall short socially.

With the initial 13-lever menu the problem is unsatisfiable: the best achievable outcome still leaves CO2 37 points over, eHANPP 27 points over, and equality 13 points under, for a residual violation of 77 points. After adding a 14th lever, carbon fee-and-dividend combined with large-scale reforestation (Section 4.2), the problem becomes satisfiable, with four equally parsimonious optimal models of eight levers each:

Dimension	Type	Baseline	Model 1	Model 2	Model 3	Model 4
co2	ceiling	172	**87**	**87**	**87**	**87**
phosphorus	ceiling	120	**95**	**95**	**95**	**95**
nitrogen	ceiling	114	**89**	**89**	**89**	**89**
blue_water	ceiling	44	44	44	44	44
ehanpp	ceiling	157	**92**	**92**	**92**	**92**
eco_footprint	ceiling	129	**94**	**94**	**94**	**94**
material_footprint	ceiling	141	**96**	**96**	**96**	**96**
life_satisfaction	floor	120	**142**	**142**	**142**	**142**
healthy_life_exp	floor	117	117	117	117	117
nutrition	floor	120	120	120	**110**	**110**
sanitation	floor	98	**113**	**113**	**123**	**123**
income	floor	104	**134**	**134**	**114**	**114**
energy_access	floor	105	**120**	**120**	**120**	**120**
education	floor	112	112	112	112	112
social_support	floor	99	**104**	**109**	**109**	**104**
democratic_quality	floor	99	**119**	**129**	**129**	**119**
equality	floor	62	**102**	**102**	**102**	**102**
employment	floor	93	**103**	**103**	**103**	**103**

All four models share the same six core levers, `l1_renewable_energy, l2_circular_economy, l3_sustainable_agriculture, l5_working_time_reduction, l9_wealth_tax, l14_carbon_fee_dividend_reforestation`, and differ only in which two of `l6_universal_basic_income`, `l8_participatory_governance`, `l12_decentralization_reform`, `l13_sanitation_infrastructure` close the remaining social gaps: Model 1 combines the core with L6 and L12, Model 2 with L6 and L8, Model 3 with L8 and L13, Model 4 with L12 and L13. None of these four levers touches an ecological dimension, so all seven ecological values are identical across models; `nutrition`, `sanitation` and `income` shift between L6 and L13, while `social_support` and `democratic_quality` shift between L8 and L12.

This is the clearest illustration in the project of the model surfacing a genuine value choice. Several institutional designs are formally equivalent with respect to the doughnut's constraints, but they are not equivalent in what kind of society they build: participatory governance and top-down decentralization reflect different readings of democratic legitimacy, and the ASP encoding correctly leaves that choice to human judgement rather than resolving it computationally.

7.3 United States: structurally infeasible

In the baseline, all seven ecological dimensions are in overshoot, several of them massively so: CO2 at 1314, more than 13 times the sustainable share; phosphorus at 782; nitrogen at 663; material footprint at 378; ecological footprint at 393. Only equality (81) and employment (88) fall short socially, and only slightly.

Dimension	Type	Baseline	Best-effort result	Status
co2	ceiling	1314	1229	still **1129** over
phosphorus	ceiling	782	757	still **657** over
nitrogen	ceiling	663	638	still **538** over
eco_footprint	ceiling	393	358	still **258** over
material_footprint	ceiling	378	333	still **233** over
blue_water	ceiling	106	86	met
ehanpp	ceiling	142	77	met
life_satisfaction	floor	117	139	met
healthy_life_exp	floor	117	117	met
nutrition	floor	194	184	met
sanitation	floor	105	105	met

Dimension	Type	Baseline	Best-effort result	Status
income	floor	107	117	met
energy_access	floor	106	121	met
education	floor	100	120	met
social_support	floor	104	104	met
democratic_quality	floor	102	102	met
equality	floor	81	121	met
employment	floor	88	108	met

Even with all 14 levers available, the problem remains unsatisfiable. The best-effort diagnostic selects eight levers, `l1_renewable_energy, l2_circular_economy, l3_sustainable_agriculture, l4_water_efficiency, l5_working_time_reduction, l9_wealth_tax, l10_education_investment, l14_carbon_fee_dividend_reforestation`, the largest and most varied lever set of the three case studies. This fully closes the social floor, with all 11 dimensions met, but leaves five of the seven ecological ceilings breached, for a total residual violation of 2815 points (the five overshoot figures sum exactly: 1129 + 657 + 538 + 258 + 233), roughly 36 times Costa Rica's pre-fix residual. `blue_water` and `ehanpp` are brought under their ceilings; the impossibility concentrates specifically in CO2, phosphorus and nitrogen, where US overshoot is most extreme to begin with.

This is, in formal terms, exactly the empirical finding of O'Neill et al. (2018) themselves, that no country in their dataset meets the social floor without transgressing the ecological ceiling, reproduced here as a logical consequence of the model rather than a restatement of the paper's regression results. Given real baseline data and levers bounded to plausible real-world magnitudes (Section 5), no subset of marginal reforms is enough at the scale of US overshoot. This operationalizes, computationally, the central argument of *Doughnut Economics* against incremental reform of high-consumption economies.

8. Discussion and limitations

- **Effect sizes.** These are assumptions, not estimates. The representativeness check in Section 5 bounds them against real cross-country variation, but it cannot make them causal. A different, equally defensible set of magnitudes could shift Costa Rica back to unsatisfiable, or shift the United States implausibly to satisfiable. What survives this uncertainty is the ordering: the United States' residual violation, 2815, is roughly two orders of magnitude larger than Costa Rica's pre-fix residual of 77, and that ordering is far more robust to the specific numbers chosen than any single feasibility verdict is.
- **Static model.** The model compares one baseline year against one post-reform state, and says nothing about transition dynamics, the time each lever takes to produce its effect, or path-dependency between levers, for instance whether renewable investment becomes cheaper once a circular-economy policy has already reduced material demand. A dynamic extension would need a temporal ASP formulation or a hybrid with system-dynamics simulation.
- **Additive lever effects.** The encoding sums each selected lever's effect on a dimension with no interaction terms, while in reality combined policies plausibly have synergies or redundancies, since two levers both targeting material throughput may not simply add. This simplification is what keeps the best-effort diagnostic well defined and fast to compute, at the cost of realism.
- **Best-effort as a stand-in for a minimal core.** The diagnostic model is a substitute for a true unsatisfiable core, chosen deliberately (Section 4.4) because it reports the explanation in units an ethical argument can use, namely remaining floor or ceiling violation, rather than a set of conflicting logic-program rules that would be harder to interpret substantively. A genuine minimal-core extraction, via `clingo`'s assumption-based core computation, was considered and set aside for this project given the added complexity of handling the `conflicts/2` mutual-exclusion constraint correctly inside an assumption set; it is noted here as a natural next step rather than a hidden gap.
- **Country-level aggregation.** Both the floor and ceiling indicators are national averages, so a country meeting the income floor on average can still have a large share of its population below it. The model inherits this limitation directly from the underlying O'Neill et al. dataset.

9. Conclusion

Framing Doughnut Economics as an ASP constraint-satisfaction problem over real country data does two things that a purely descriptive or purely optimization-based treatment would not. First, it forces every implicit ethical commitment into the open: treating both floor and ceiling as hard constraints, rather than as terms in a weighted objective, is a direct, auditable encoding of sufficientarianism and strong sustainability, not an incidental modeling convenience. Second, when the problem is infeasible the model does not simply fail silently or report an infeasibility flag: it produces a structured, quantified account of which constraints remain unreachable and by how much, which is what turns a solver result into something a policy discussion can actually use.

Applied to real data, the model reproduces the central empirical claim of Doughnut Economics as a logical consequence of its own stated assumptions rather than as an assertion: at the scale of overshoot displayed by a country like the United States, no combination of the kind of marginal policy levers considered here is sufficient. At the same time, the model shows this is not a universal verdict. Vietnam's situation is comfortably solvable, and Costa Rica sits in between, solvable only with a more ambitious lever and, even then, through several institutionally distinct and formally equivalent paths whose final choice the model correctly leaves to human political judgement rather than resolving computationally.

References

- Autonomy / Alda — Haraldsson, G. D., & Kellam, J. (2021). *Going Public: Iceland's Journey to a Shorter Working Week.* Autonomy / Alda.
- Ellen MacArthur Foundation (2013). *Towards the Circular Economy: Economic and Business Rationale for an Accelerated Transition.*
- Frankfurt, H. (1987). Equality as a Moral Ideal. *Ethics*, 98(1), 21–43.
- Hansen, J. et al. (2013). Assessing "Dangerous Climate Change": Required Reduction of Carbon Emissions to Protect Young People, Future Generations and Nature. *PLOS ONE*.
- IEA (2021). *Net Zero by 2050: A Roadmap for the Global Energy Sector.* International Energy Agency.
- ILO (2012). *Social Protection Floors Recommendation, 2012 (No. 202).* International Labour Organization.
- IPCC (2022). *Climate Change 2022: Mitigation of Climate Change.* Working Group III Contribution to AR6.
- Kallis, G., Kostakis, V., Lange, S., Muraca, B., Paulson, S., & Schmelzer, M. (2018). Research on Degrowth. *Annual Review of Environment and Resources*, 43, 291–316.
- Nussbaum, M. (2011). *Creating Capabilities: The Human Development Approach.* Harvard University Press.
- OECD (2020). *Innovative Citizen Participation and New Democratic Institutions: Catching the Deliberative Wave.* OECD Publishing.
- O'Neill, D. W., Fanning, A. L., Lamb, W. F., & Steinberger, J. K. (2018). A good life for all within planetary boundaries. *Nature Sustainability*, 1, 88–95.
- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press.
- Piketty, T. (2014). *Capital in the Twenty-First Century* (A. Goldhammer, Trans.). Harvard University Press.
- Raworth, K. (2017). *Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist.* Chelsea Green Publishing.
- Rockström, J. et al. (2009). A safe operating space for humanity. *Nature*, 461, 472–475.
- Sen, A. (1999). *Development as Freedom.* Oxford University Press.
- UNESCO (2015). *Incheon Declaration and Framework for Action for the Implementation of Sustainable Development Goal 4.*
- UN-Water (2021). *The United Nations World Water Development Report 2021: Valuing Water.* UNESCO, on behalf of UN-Water.
- WHO & World Bank (2017). *Tracking Universal Health Coverage: 2017 Global Monitoring Report.*
- WHO/UNICEF Joint Monitoring Programme (2021). *Progress on Household Drinking Water, Sanitation and Hygiene 2000–2020: Five Years into the SDGs.*
- Willett, W. et al. (2019). Food in the Anthropocene: the EAT–Lancet Commission on healthy diets from sustainable food systems. *The Lancet*, 393(10170), 447–492.